



DATAFACTZ AI ENGINEERING INTERNSHIP · WEEK 2, USE CASE 2

Document Extraction System

A human-in-the-loop invoice extraction pipeline —
parse, extract, score, escalate, review, export.

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THE PROBLEM

Manual invoice entry doesn't scale — full LLM trust isn't safe either

Manual entry

Slow, repetitive, error-prone at volume. Review cost is the same whether a document was trivial or genuinely ambiguous.

Unchecked LLM extraction

A model can misread a total or drop a line item — and report it with the same confidence as a correct extraction. Not acceptable for financial documents.

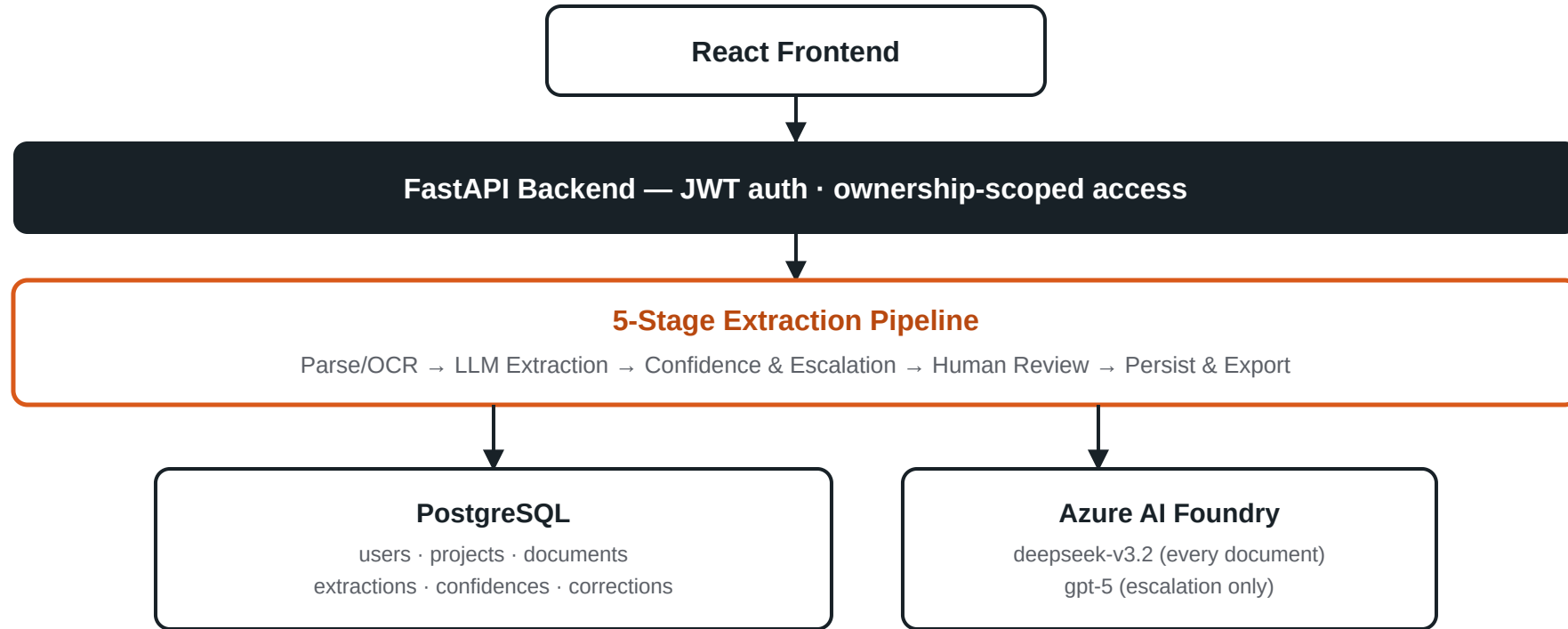
This project

Automate extraction, measure real confidence, escalate only what's uncertain, and route only genuine uncertainty to a human.

Constraints: open-source parsing/OCR only (no managed cloud document-AI) · digital PDFs and scanned images in one pipeline · corrections stored separately from raw model output, so pre- and post-review accuracy can both be measured.

ARCHITECTURE

Three tiers, one pipeline, two model deployments



Route by input type, extract, score honestly

1

Parse / OCR Routing

Text layer present → PyMuPDF.

No text layer → render at 200 DPI →

PaddleOCR, isolated in its own worker process
so it can't block the API.

2

LLM Extraction

DeepSeek-V3.2 against a Pydantic schema.

Invalid JSON gets exactly one repair-prompt
retry, quoting the validation error — never
crashes, never loops.

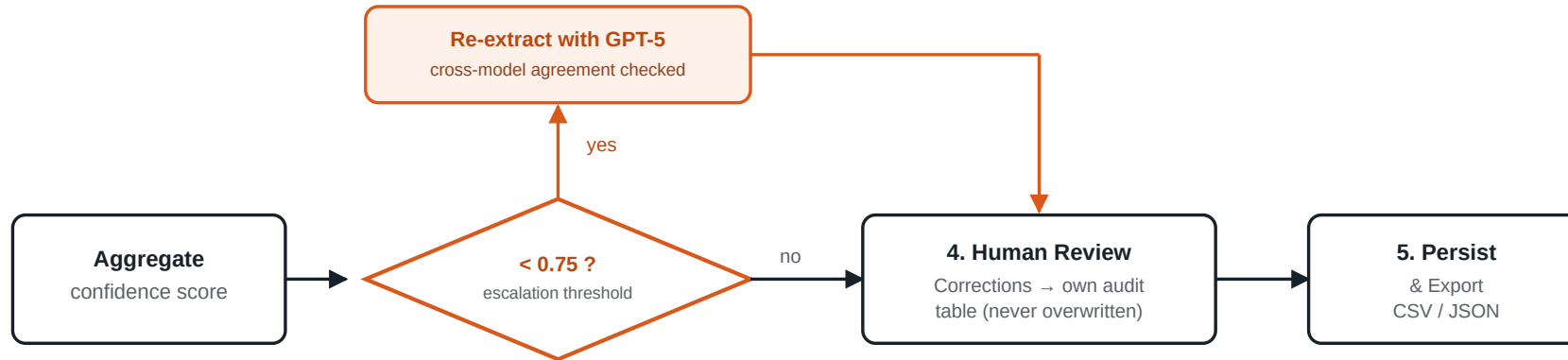
3

Confidence Scoring

$\text{composite} = 0.5 \times \text{self-reported} + 0.5 \times \text{heuristic}$ — format checks,
subtotal/total reconciliation, and OCR-quality
discount. Not the model's raw self-report.

Every document runs stages 1–3. What happens next depends on the score.

Only uncertain documents cost more or take longer

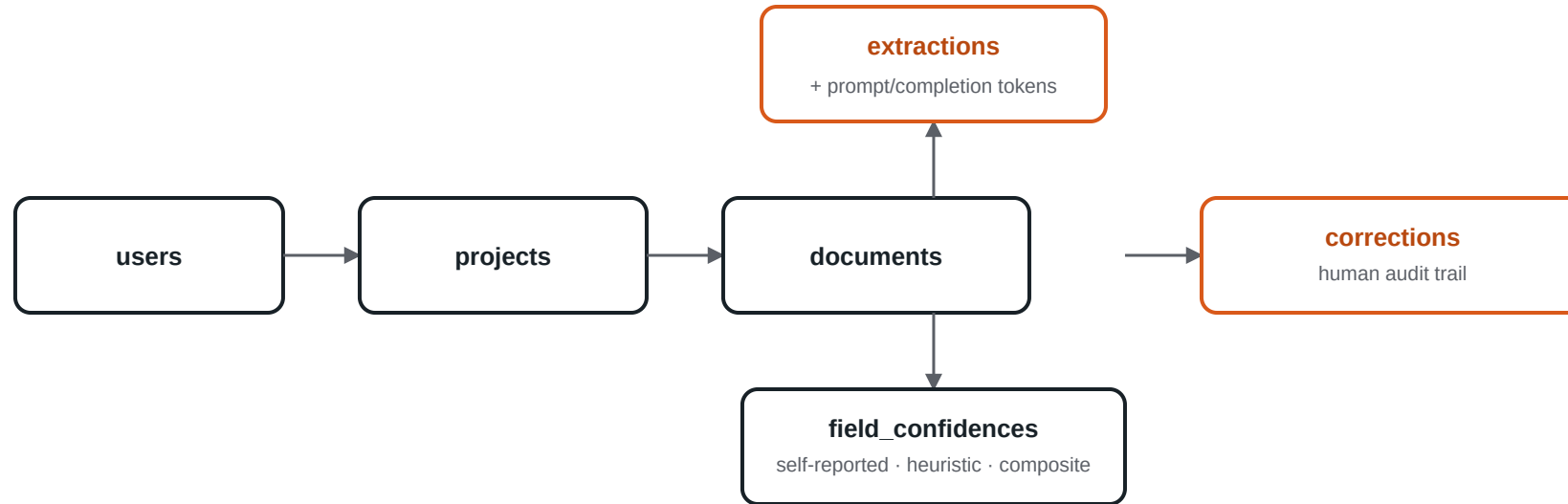


Below threshold: re-extracted with GPT-5; cross-model agreement computed only on this subset — running both models on every document was rejected as doubling cost with no escalation benefit.

Every document — escalated or not — still reaches a human reviewer. Confidence flags which fields need a closer look; it never skips review outright.

DATA MODEL

Raw extraction and human corrections are never the same row



Kept separate on purpose: the accuracy report compares **raw-model accuracy** against **post-review accuracy** from the same stored history — collapsing them into one "final record" table would make that comparison impossible.

KEY DESIGN DECISIONS

Four choices, and what was rejected

Two-tier escalation

DeepSeek on every doc, GPT-5 only below threshold.

Rejected: running both on every document — doubles cost, no benefit.

Composite confidence

Self-reported × heuristic checks.

Rejected: raw self-reported score alone — models are often overconfident.

Always human-reviewed

No confidence-based auto-approval, ever.

Rejected: auto-approve above threshold — one wrong total is worse than one extra review.

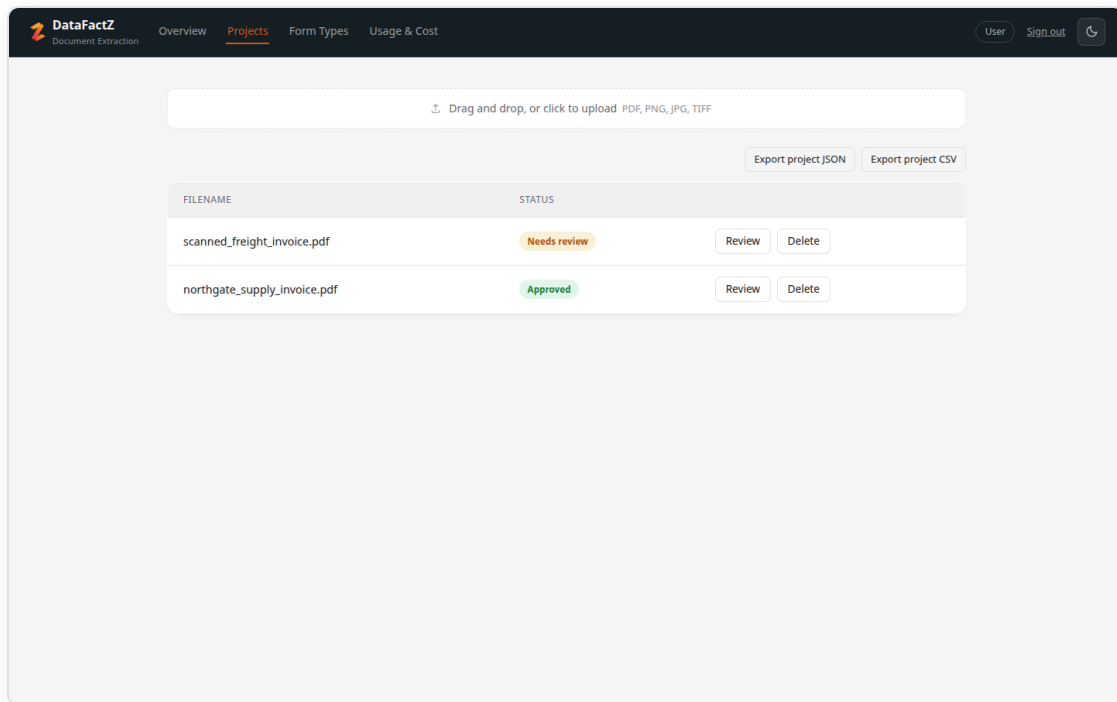
Cascade-delete on user removal

Shipped, then revised: orphaning a deleted user's projects onto admin was live first.

Replaced once it was clear that handed admin someone else's reviewed documents by default.

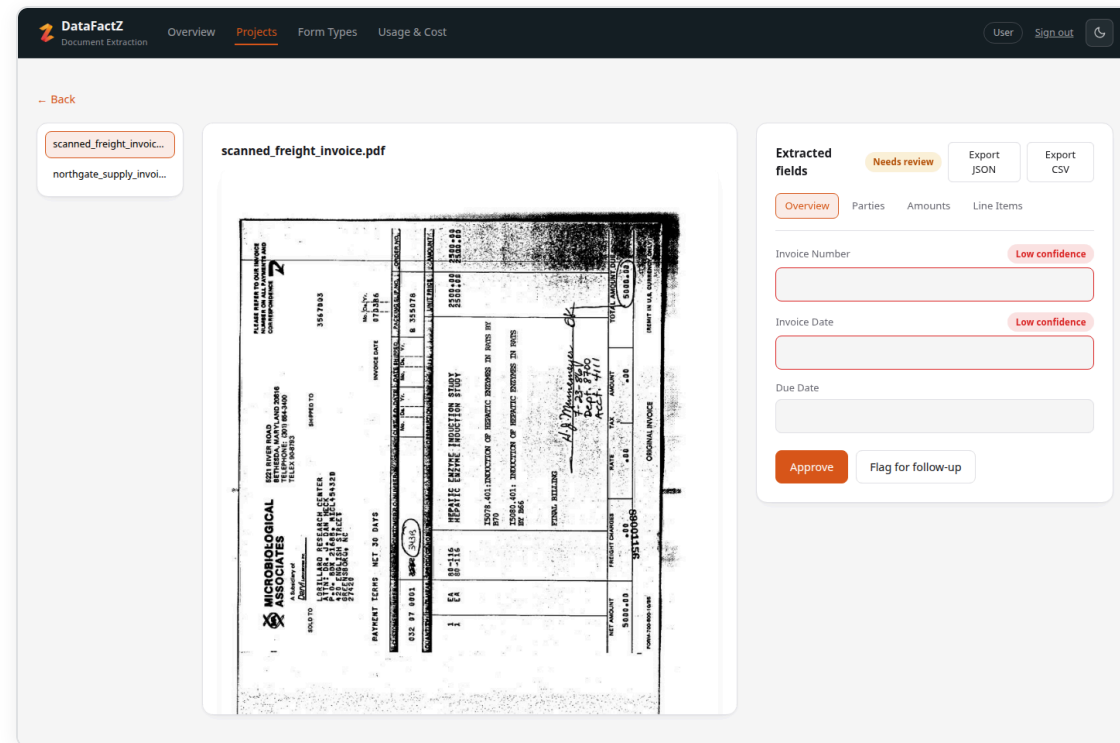
WORKING DEMO

Real run: one clean digital invoice, one genuinely illegible scan



The screenshot shows the DataFactZ Document Extraction interface. At the top, there's a navigation bar with 'Overview', 'Projects', 'Form Types', and 'Usage & Cost'. Below this, a large area for uploading documents is visible, with a prompt 'Drag and drop, or click to upload PDF, PNG, JPG, TIFF'. A table lists the uploaded documents:

FILENAME	STATUS	Actions
scanned_freight_invoice.pdf	Needs review	Review, Delete
northgate_supply_invoice.pdf	Approved	Review, Delete



The screenshot shows the DataFactZ interface with the 'scanned_freight_invoice.pdf' document selected. The document is displayed as a scanned image. On the right, the 'Extracted fields' panel shows the following information:

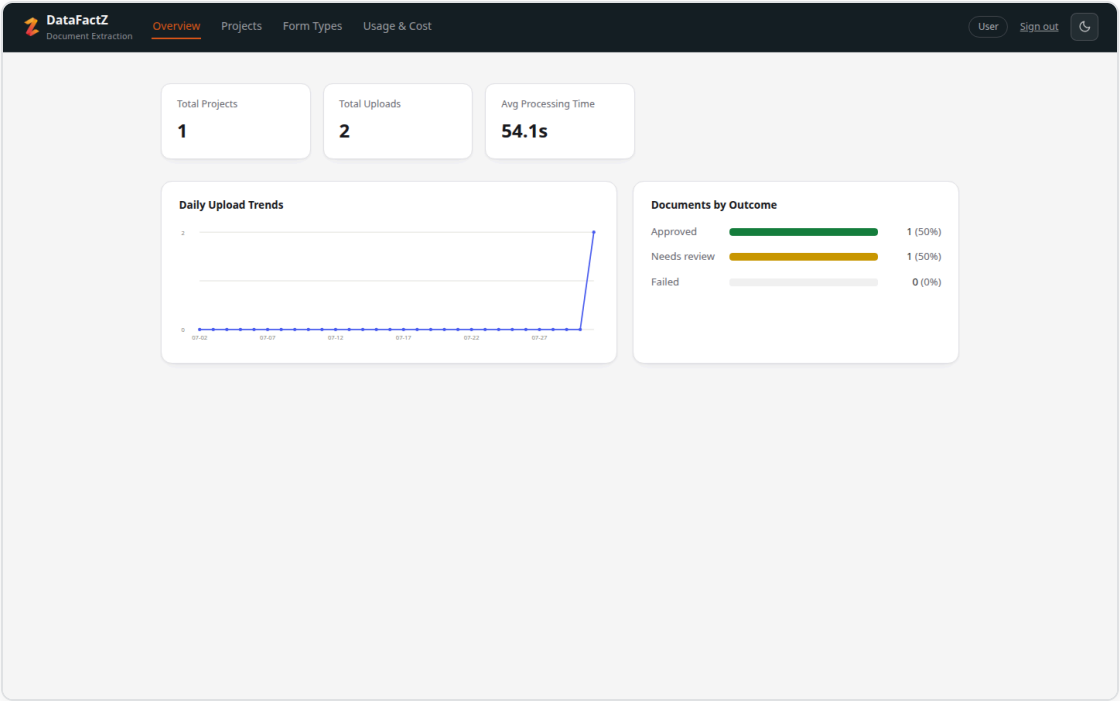
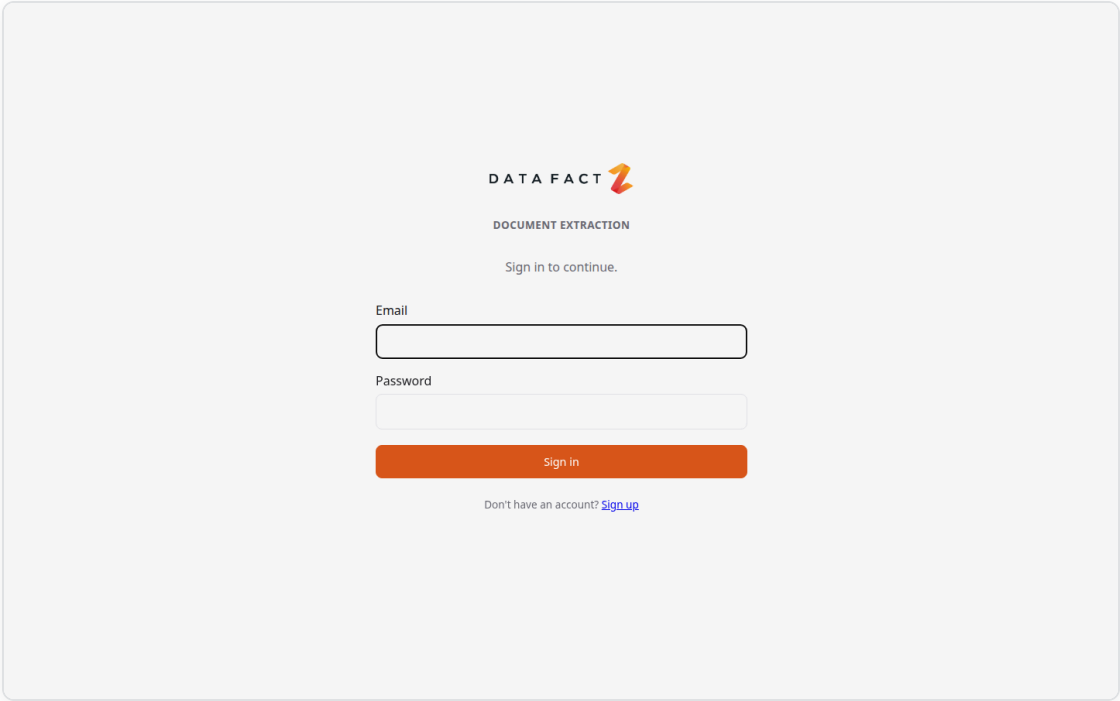
- Invoice Number: [Redacted] (Low confidence)
- Invoice Date: [Redacted] (Low confidence)
- Due Date: [Redacted]

The 'Extracted fields' panel also includes buttons for 'Needs review', 'Export JSON', and 'Export CSV'. Below the fields, there are buttons for 'Approve' and 'Flag for follow-up'.

The scanned invoice escalated to GPT-5 and still came back illegible on every field — composite score 0.25 — exactly the case the pipeline is built to hand to a human instead of guessing at.

BRANDED UI

DataFactZ navy & orange, light and dark theme



COST ESTIMATE

Measured, not guessed — real token counts from this deployment

\$0.0015

non-escalated document
(DeepSeek only)

\$0.0402

escalated document
(DeepSeek + GPT-5)

~28×

cost multiplier when
a document escalates

100 users

~2,000 docs/mo → est. **\$15–\$34/mo** LLM cost at 15–40% escalation, plus
~\$45–75/mo infra.

5,000 users

~100,000 docs/mo → est. **\$730–\$1,700/mo** LLM cost at 15–40% escalation,
plus ~\$700–1,300/mo infra.

BUILT WITH CLAUDE CODE

Every commit is AI-paired — design, implementation, and this deck

How it was used

Design discussion before implementation on every major decision; real bug diagnosis (OCR blocking the event loop, a delete-user FK ordering bug, a schema-validation gap); this report and deck generated from the live system and real git history, not templated text.

What's next at scale

Move pipeline execution to a Celery/Redis queue; run the OCR worker pool on dedicated CPU instances; stateless API replicas behind a load balancer with Postgres connection pooling.



Thank you

Questions & discussion